

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

WAVE MOTION

LECTURE-01 **PHYSICS**

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Topics To Be Covered

- 1** Wave Motion
- 2** Transverse Waves
- 3** Longitudinal Waves
- 4** Speed of travelling wave
- 5** Displacement Relation for a Progressive Wave

Waves

Wave motion: Transverse and longitudinal waves, speed of travelling
ave/displacement relation for a progressive wave principle of
superposition of waves, reflection of waves, ^{* 5 mark} ~~standing~~ waves in strings
and organ pipes, fundamental mode and harmonics, Beats.

lecture . no . 1

①

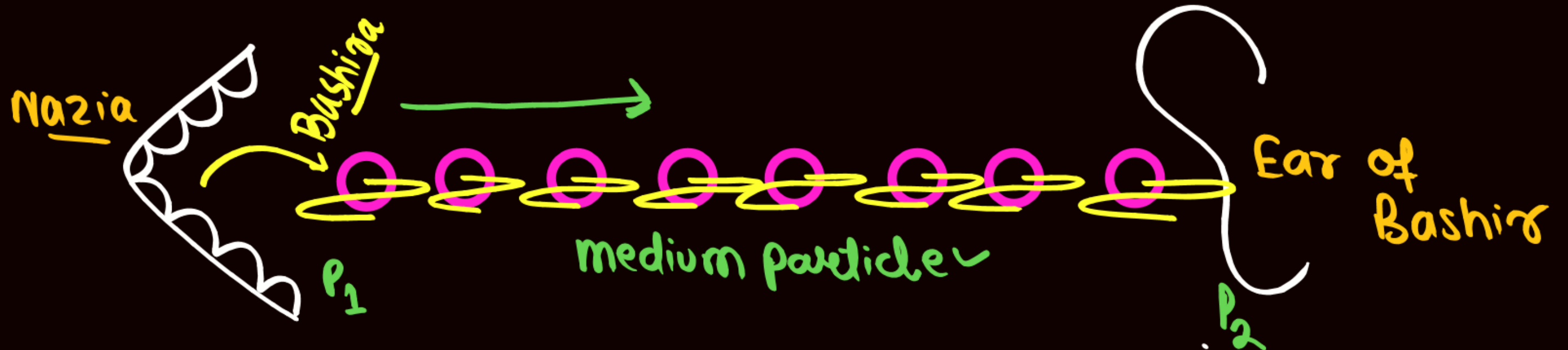
wave motion

- * vibrational motion
- * Too and forth motion
- * oscillation motion.

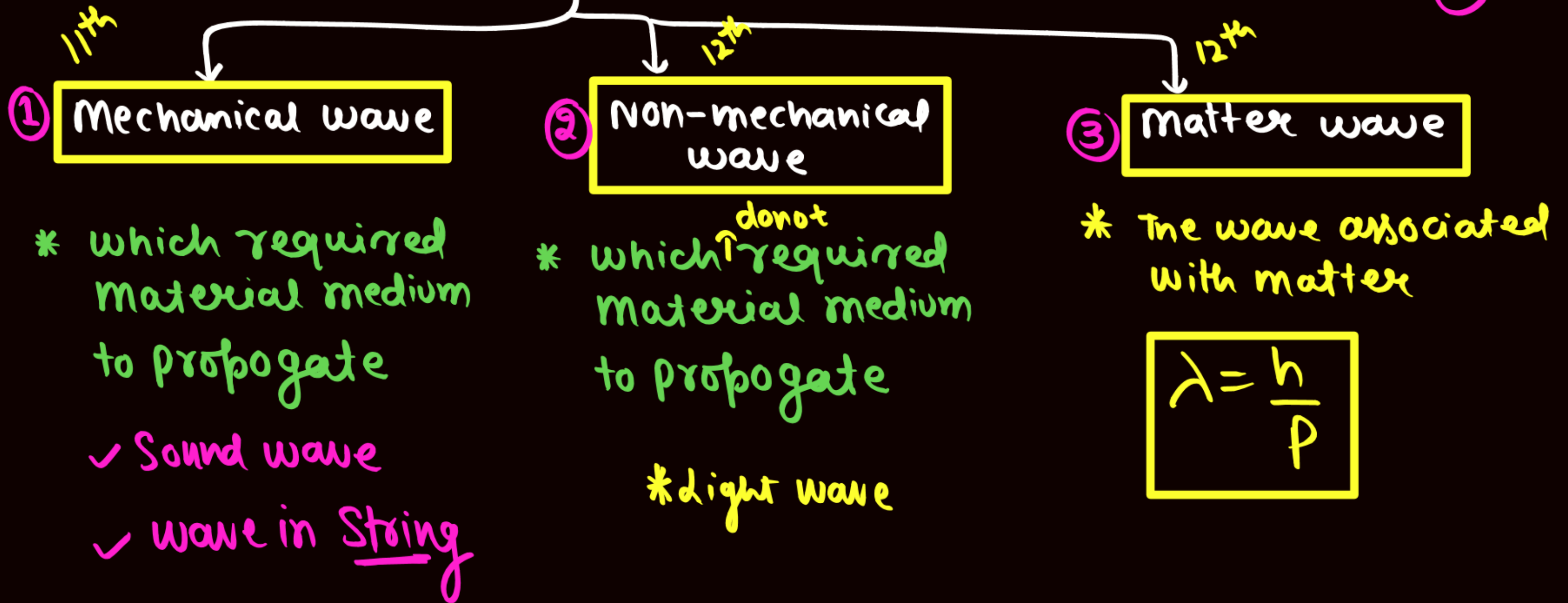


Energy

wave  wave  is the disturbance that propagate from one point to another point



wave

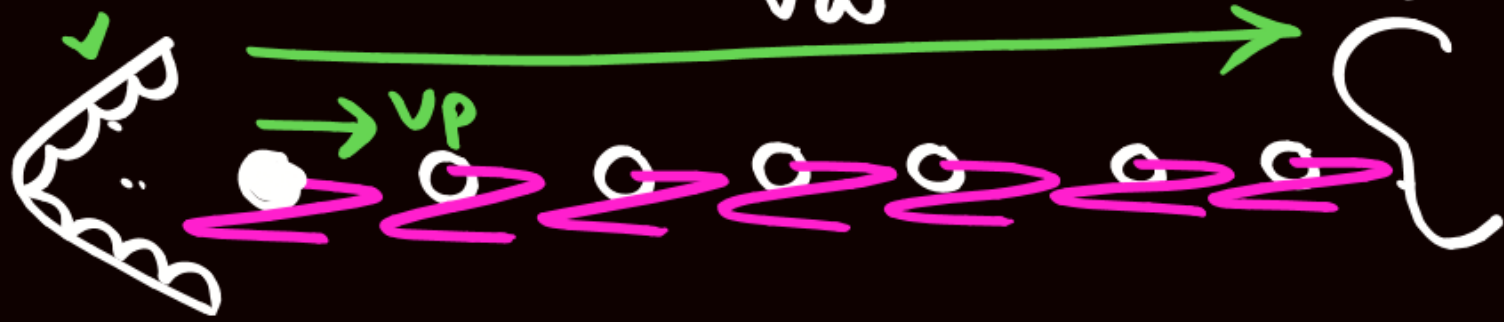


Mechanical wave



① Longitudinal wave

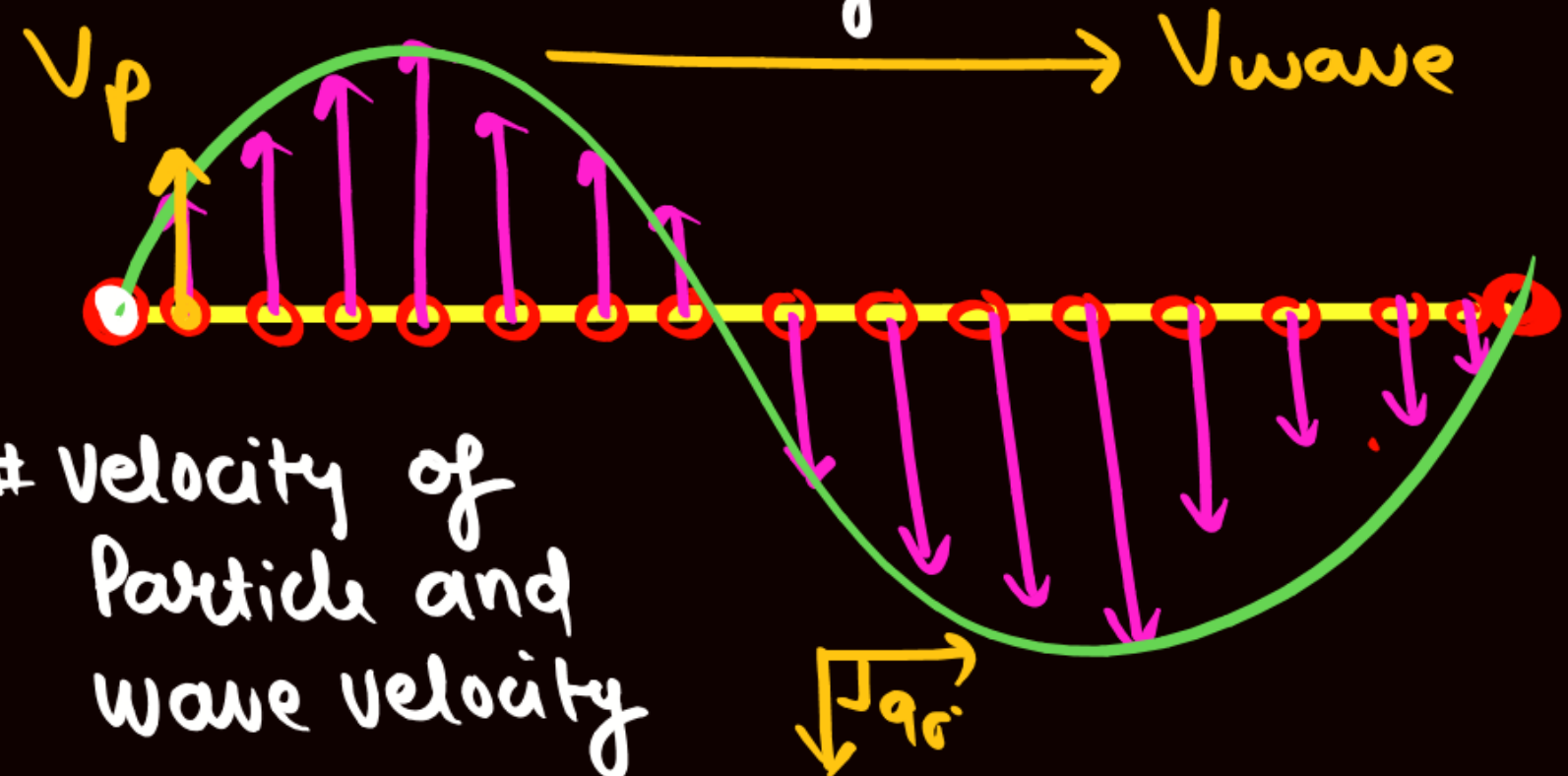
- Sound wave
- V_s → velocity of wave Energy



In longitudinal wave
Particle velocity and wave
velocity are along the
Same line

② Transverse wave

- wave in a string



Velocity of
Particle and
wave velocity
Both are
Perpendicular to each other

Plane progressive wave 3M-2024

④

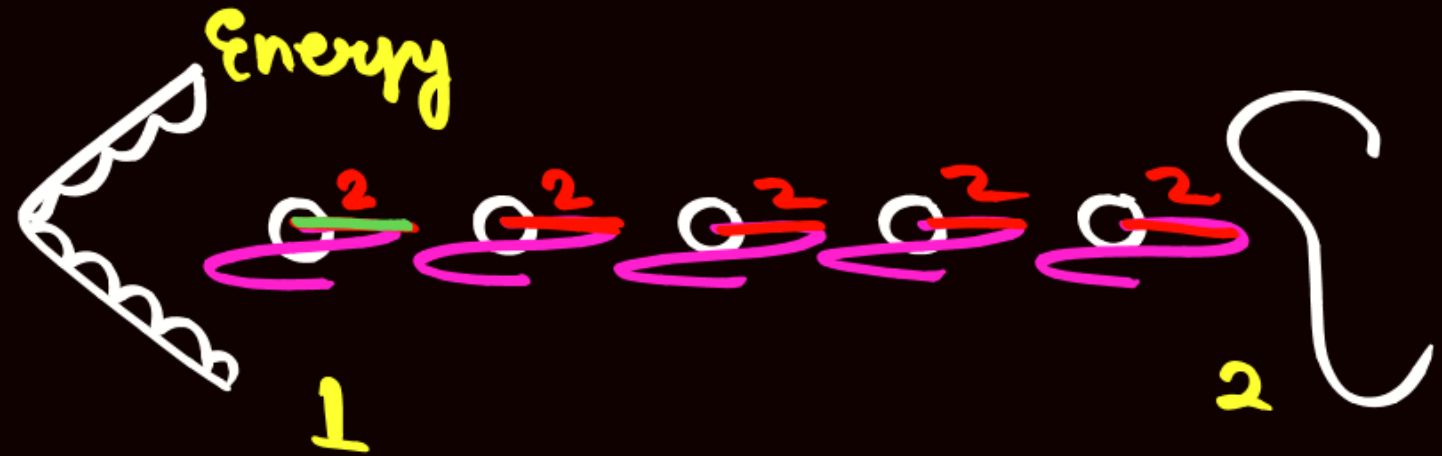
① The wave which propagate in a medium with constant amplitude

② no. particle is at rest

③ all particle does SHM

④ all particle have different phase

⑤ Plane progressive wave undergoes Reflection
Refraction

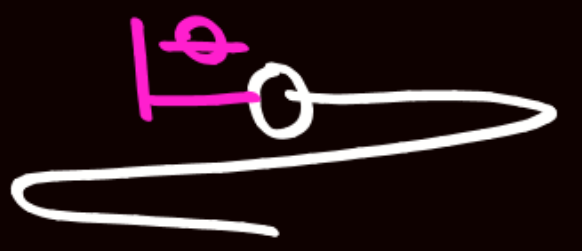
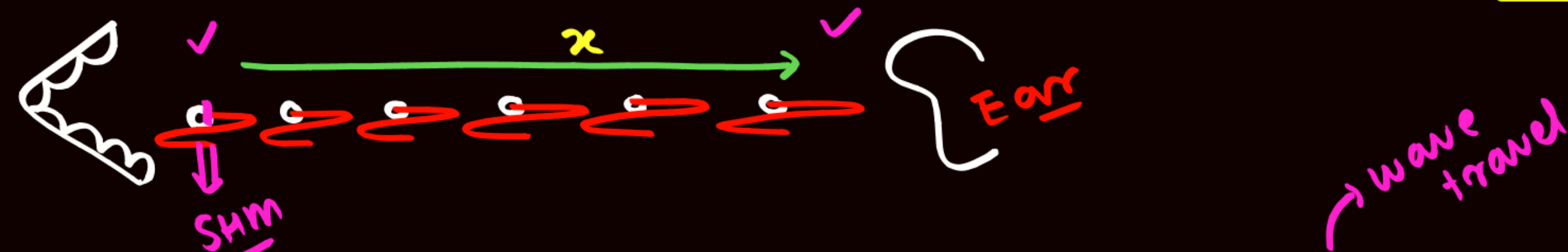


Equation of plane progressive wave

(no derivation)

- The wave which propagate in medium with constant **Amplitude**

⑤



$K \sim$ Propagation constant.

$$K = \frac{\omega}{v}$$

$\omega \rightarrow$ angular velocity
 $v \rightarrow$ wave velocity

$$K = \frac{2\pi}{\lambda}$$

$\lambda \rightarrow$ wave length

$$x = A \sin(\omega t - Kx)$$

$$x = A \sin(\omega t - \frac{\omega}{v}x)$$

$$x = A \sin(\omega t - \frac{2\pi}{\lambda}x)$$

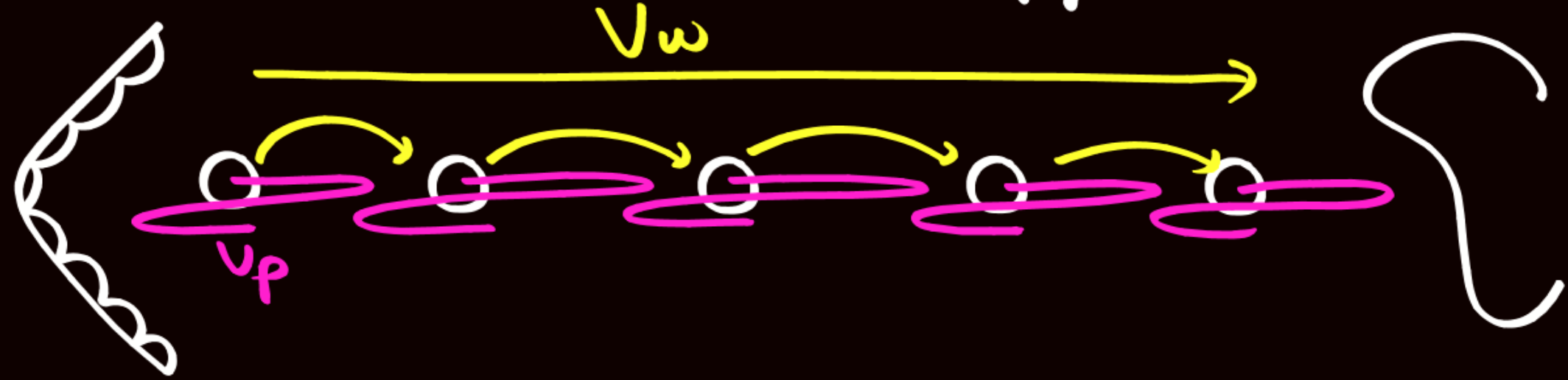
$$x = A \sin(\omega t - Kx + \phi)$$

$\phi \rightarrow$ if there is phase difference

Wave Velocity V_w Energy

⑥

the velocity with which the wave energy moves in a medium



Wave
velocity

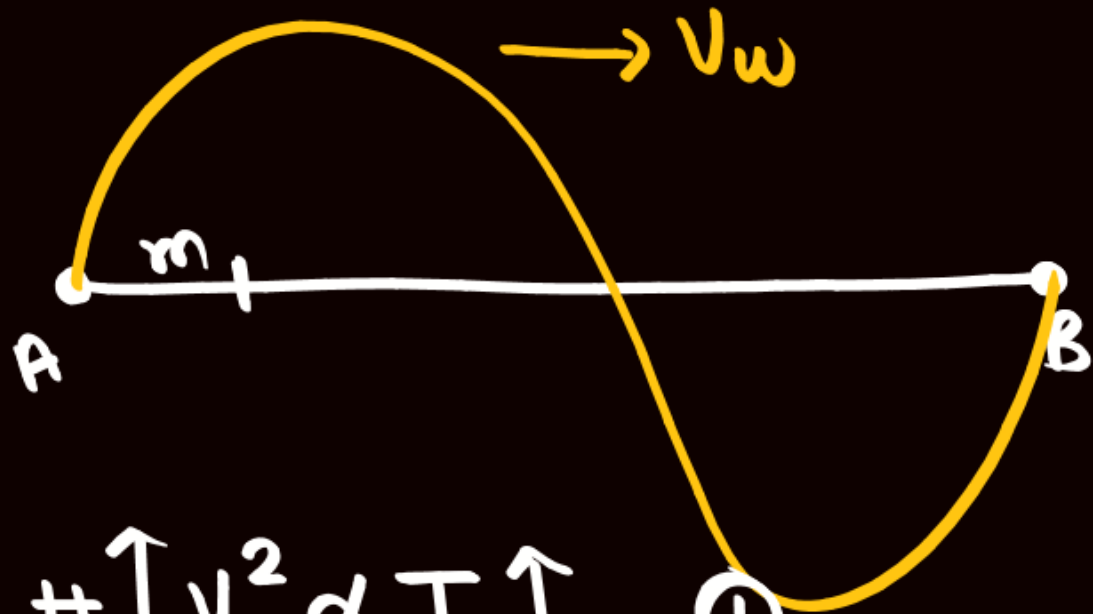
$$k = \frac{\omega}{v}$$

$$v = \frac{\omega}{k}$$

Velocity of wave in stretched string

/ Transverse wave

7



$$\# \uparrow v^2 \propto T \uparrow - \textcircled{\text{I}}$$

$$\downarrow v^2 \propto \frac{1}{\mu} \uparrow - \textcircled{\text{II}}$$

$$v^2 \propto \frac{T}{\mu}$$

$$v^2 = K \frac{T}{\mu}$$

$$v^2 = 1 \frac{T}{\mu}$$

$K \Rightarrow$ Constant

$$K=1$$

$$v^2 = \frac{T}{\mu}$$

$$v = \sqrt{\frac{T}{\mu}}$$



$$v_w = \sqrt{\frac{T}{\mu}}$$

Tension

mass per unit length

$$\mu = \frac{m}{L}$$

$$v = \sqrt{\frac{T}{m/L}}$$

$$v = \sqrt{\frac{TL}{m}}$$



Velocity of sound wave (longitudinal wave)

* Newton



$$\uparrow v^2 \propto p \uparrow (\text{Pressure}) \quad \text{--- ①}$$

$$\downarrow v^2 \propto \frac{1}{\rho} (\text{density}) \uparrow \rho = \frac{m}{v} \uparrow \quad \text{--- ②}$$

$$v^2 \propto \frac{p}{\rho}$$

$$v^2 = K \frac{p}{\rho}$$

$K \rightarrow \text{constant}$

$$K = 1$$

$$v^2 = 1 \frac{p}{\rho}$$

$$v^2 = \frac{p}{\rho}$$

$$v = \sqrt{\frac{p}{\rho}}$$

$$v = \sqrt{\frac{p}{\rho}}$$

$$P = 10^5 \text{ pascal}$$

$$\rho = 1.23$$

$$v = \sqrt{\frac{10^5}{1.23}}$$

$$v = 280 \text{ m/s}$$

$$v = 331 \text{ m/s}$$

~ correction ~

SUMMARY

WM

W

3

M

2

PPW

Eq

V

✓

V

✓

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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Thank
You

